

1 Supplementary Materials

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34 **1 - Summary Tables**

35 **1-A: Age Summary Statistics**

	Statistic	Value
	N	1643.00
	Mean	61.95
36	Standard Deviation	11.47
	Median	64.00
	First Quartile (Q1)	55.00
	Third Quartile (Q3)	70.00

TABLE 1 Overall Age Summary

37 **1-B: Age by Gender Summary Statistics**

	Gender	N (%)	Mean	Standard Deviation	Median	Q1	Q3
38	female	671 (40.8%)	62.27	11.37	64	56	70
	male	972 (59.2%)	61.73	11.54	64	55	70

TABLE 2 Age Summary by Gender

39 **1-C: Age by Decade Summary Statistics**

40	Age (Decade)	N (%)
	20–29	22 (1.3%)
	30–39	46 (2.8%)
	40–49	164 (10.0%)
	50–59	356 (21.7%)
	60–69	616 (37.5%)
	70–79	391 (23.8%)
	80–89	47 (2.9%)
	90–99	1 (0.1%)

TABLE 3 Age Summary by Decade

41 **1-D: Country Counts and Proportions**

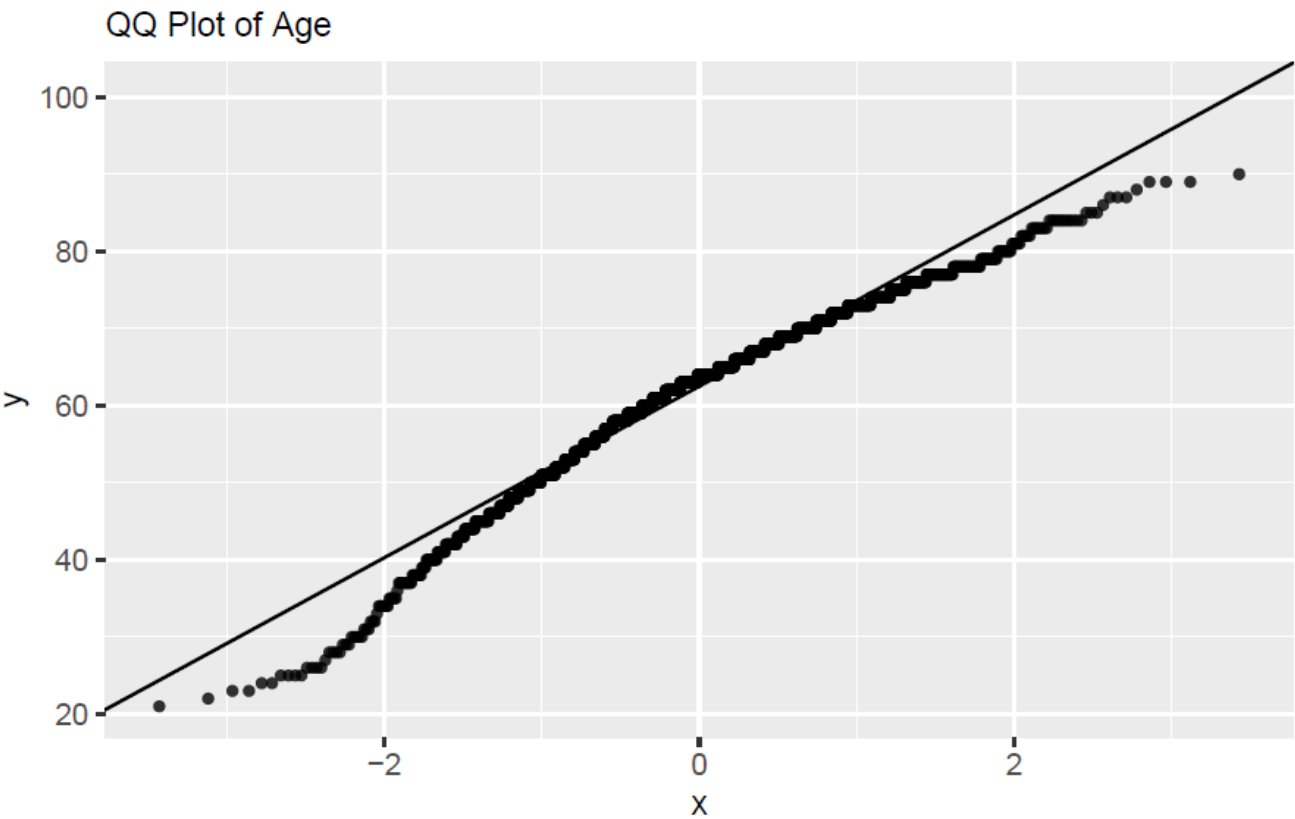
42

Country	N (%)
AUT	154 (9.4%)
CAN	27 (1.6%)
CHN	128 (7.8%)
DEU	125 (7.6%)
FRA	156 (9.5%)
IND	60 (3.7%)
ITA	139 (8.5%)
JPN	696 (42.4%)
USA	158 (9.6%)

TABLE 4 Country Overall Summary (NATO country code, trigram)

43 2 - Visualizations

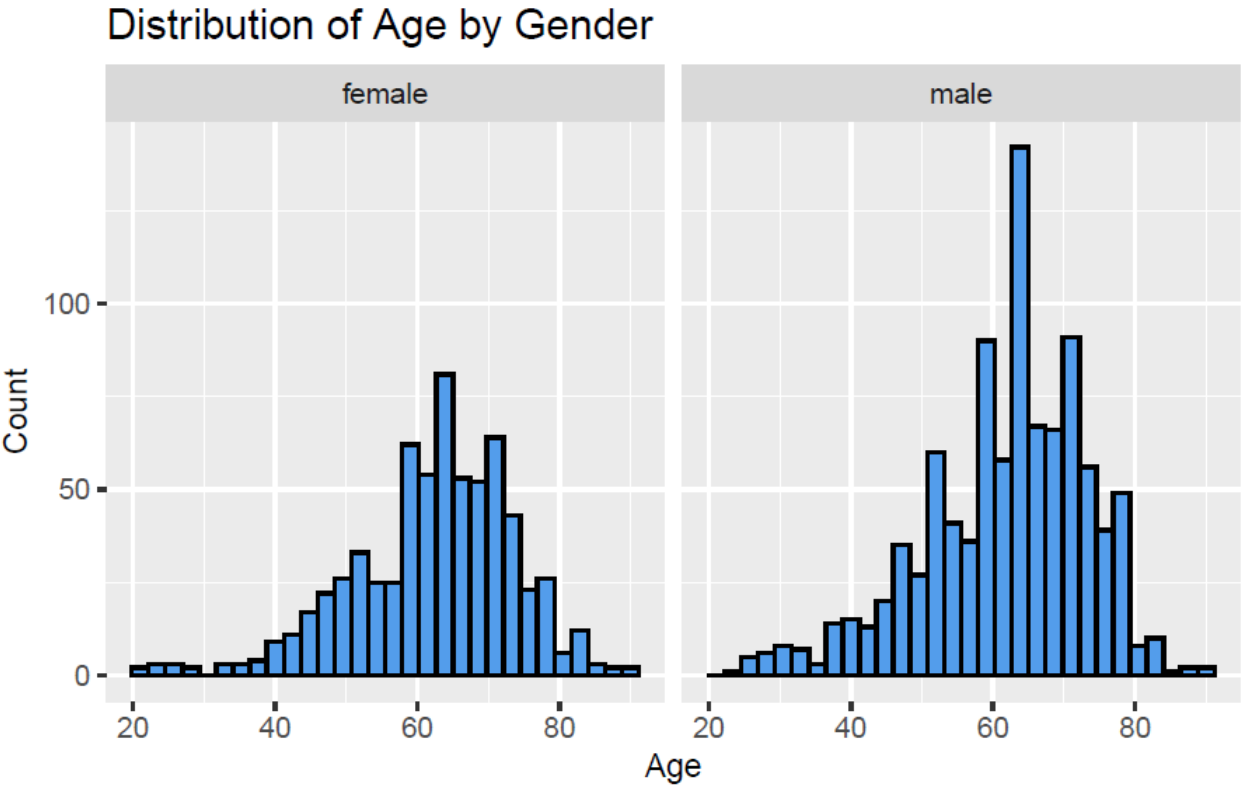
44 2-A: Age QQ-Plot



45

FIG 1 QQ Plot supporting histogram displaying normality distribution of age in the dataset.

46 2-B: Age Histogram by Gender



47

FIG 2 Histogram of age, faceted by gender.

48 2-C: Country Distribution and Proportions

49

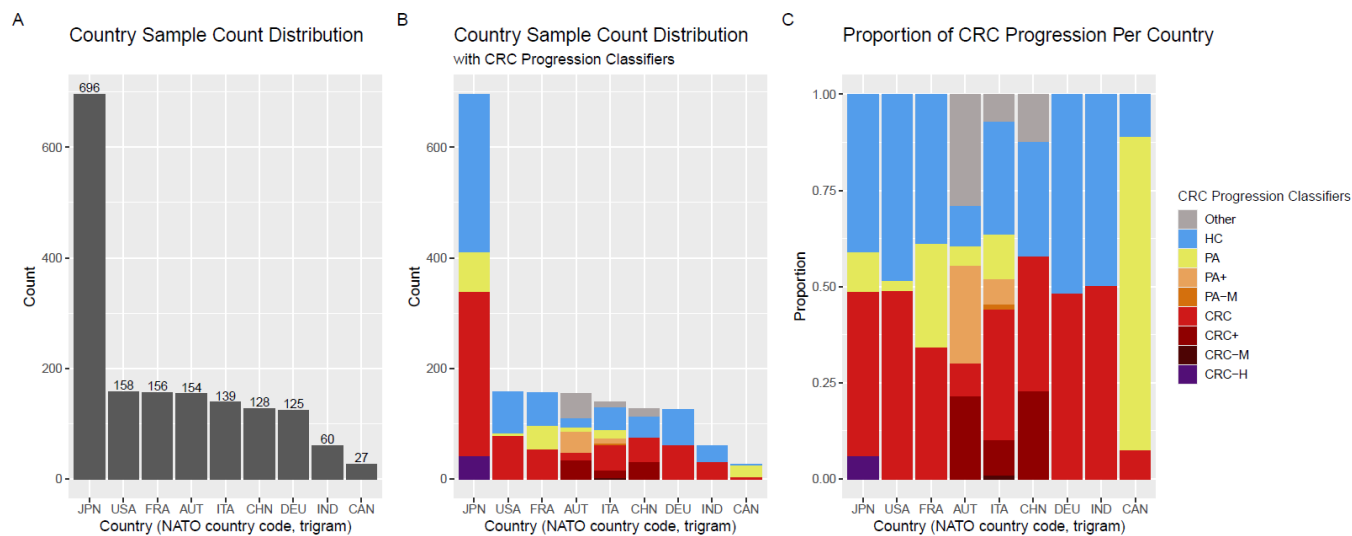
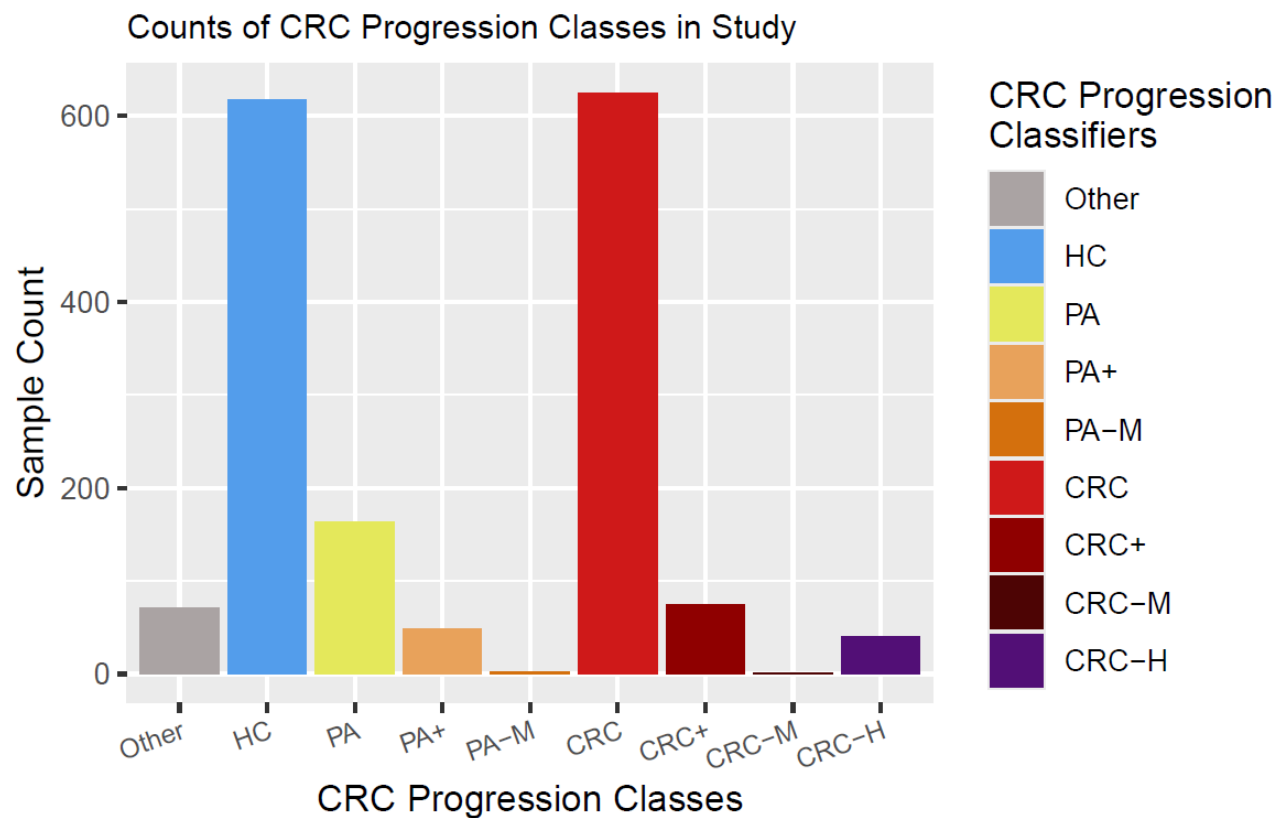


FIG 3 (A) Bar plot of country sample count, sorted in descending order. (B) Stacked bar plot of county sample count, sorted in descending order, filled with CRC progression classifiers. (C) Stacked Bar plot of the CRC progression classifier proportion for each country in the multi-cohort study.

50 2-D: CRC Progression Classifier Counts



51

FIG 4 Barplot of CRC progression classifier counts.

52 3 - Statistical Analyses

53 3-A: CRC Progression Classifier and Age ANOVA Significant Tukey's HSD

54 Pairings

	Disease Pairing	95% CI Lower	95% CI Upper	P Value
	HC-Other	-10.50284389	-1.7236952	0.0005460
	PA+-HC	1.52805911	12.0253580	0.0020679
55	CRC-HC	1.26185019	5.2375002	0.0000151
	CRC+-HC	3.25756254	11.8241879	0.0000019
	CRC+-CRC	0.01085945	8.5715406	0.0488451
	CRC-H-CRC+	-14.81785403	-1.1021460	0.0097263

TABLE 5 Significant Tukey's HSD pairings for CRC progression and Age.

56 **3-B: CRC Progression Classifier and BMI ANOVA Significant Tukey's HSD**
57 **Pairings**

	Disease Pairing	95% CI Lower	95% CI Upper	P Value
	HC-Other	-4.94428952	-1.80320222	0.00000000
	PA-Other	-4.13977278	-0.57390021	0.0013977
	CRC-Other	-4.53643961	-1.39667564	0.00000002
	CRC-H-Other	-7.40515642	-2.45205913	0.00000000
	PA+-HC	2.37307016	6.12841806	0.00000000
58	CRC+-HC	0.73316259	3.81667022	0.0001705
	PA+-PA	1.17521532	5.29245416	0.0000411
	CRC-H-PA	-4.78350442	-0.36003813	0.0095047
	CRC-PA+	-5.72067642	-1.96643531	0.00000000
	CRC-H-PA+	-8.48746084	-3.12375118	0.00000000
	CRC+-CRC	0.32664837	3.40880795	0.0054141
	CRC-H-CRC+	-6.28817020	-1.37138642	0.0000502

TABLE 6 Significant Tukey's HSD pairings for CRC progression and BMI

59 **3-D: Linear Regression Model by CRC Progression Classifier Significant Results**

	Coefficients	Estimate	P Value
	Other	2.793178	7.00e-07
	PA	1.012194	0.004601
60	PA+	4.852519	4.19e-12
	CRC+	2.008241	0.000225
	Age centered: Other	0.158122	0.017010
	Age centered: CRC-H	0.135257	0.012499

TABLE 7 Significant Linear Regression Results for BMI and Age.

61 **4 - Hyperparameter Tuning**

62 Hyperparameter tuning was performed manually and tracked. Below is an
63 example of some of the manual tuning performed, as well as a manual check of
64 the top result from the final grid search completed using a script called
65 'Parameter_Tuning_Grid_Search.R'. The green highlighted cells indicate
66 improvement of the current cell over the baseline blue row using conditional
67 formatting.

Maximum Depth Check

train_split	prev_filter	num.trees	max_depth	mtry_fraction	min.node.size	OOB error	CM accuracy	Sensitivity	Overall F1	AUC	OFC Train	OFC Test	Overfitting Difference	Notes
0.8	0.01	500	NULL	0.01	20	0.2195541	0.6332	0.8516	0.7259953	0.69696	1	0.6332288	-0.3667712	BASELINE
0.8	0.01	500	5	0.01	20	0.2245582	0.6144	0.8571429	0.7172414	0.6695276	0.9047619	0.6144201	-0.2903418	BEST IMPROVEMENT 58
0.8	0.01	500	10	0.01	20	0.2200621	0.6458	0.8516484	0.7328605	0.6884976	1	0.645768	-0.354232	
0.8	0.01	500	15	0.01	20	0.2205077	0.6238	0.8296703	0.7156398	0.6811182	1	0.6238245	-0.3761755	
0.8	0.01	500	20	0.01	20	0.2193027	0.6238	0.8132	0.7115385	0.7031363	1	0.6238245	-0.3761755	
0.8	0.01	500	25	0.01	20	0.2197921	0.6364	0.8352	0.7238095	0.6911847	1	0.6363636	-0.3636364	

Minimum Node Size Check

train_split	prev_filter	num.trees	max_depth	mtry_fraction	min.node.size	OOB error	CM accuracy	Sensitivity	Overall F1	AUC	OFC Train	OFC Test	Overfitting Difference	Notes
0.8	0.01	500	10	0.01	NULL	0.2218224	0.627	0.8462	0.7213115	0.7007299	1	0.6269592	-0.3730408	BASELINE
0.8	0.01	500	10	0.01	5	0.220682	0.6332	0.8407	0.7234043	0.7097537	1	0.6332288	-0.3667712	Added a set.seed() here. BEST IMPROVEMENT
0.8	0.01	500	10	0.01	10	0.2220381	0.6301	0.8462	0.7230047	0.7008502	1	0.630094	-0.369906	
0.8	0.01	500	10	0.01	15	0.2202955	0.6364	0.8626	0.7302326	0.704941	1	0.6363636	-0.3636364	
0.8	0.01	500	10	0.01	20	0.2199978	0.6238	0.8242	0.7142857	0.6986043	1	0.6238245	-0.3761755	
0.8	0.01	500	10	0.01	25	0.2201942	0.6301	0.8297	0.7190476	0.7065854	1	0.630094	-0.369906	
0.8	0.01	500	10	0.01	50	0.2207809	0.605	0.8132	0.7014218	0.6851287	0.9976581	0.6050157	-0.3926424	

Mtry Fraction Check

train_split	prev_filter	num.trees	max_depth	mtry_fraction	min.node.size	OOB error	CM accuracy	Sensitivity	Overall F1	AUC	OFC Train	OFC Test	Overfitting Difference	Notes
0.8	0.01	500	10	NULL	15	0.2351554	0.5893	0.98901	0.7331976	0.665517	0.9851678	0.5893417	-0.3958261	BASELINE CM has basically NONE of the NEs correctly classified
0.8	0.01	500	10	0.01	15	0.2190599	0.6176	0.8242	0.7109005	0.6937114	1	0.6175549	-0.3824451	BEST IMPROVEMENT
0.8	0.01	500	10	0.03	15	0.2166828	0.6426	0.8352	0.7272727	0.7144461	1	0.6426332	-0.3573668	
0.8	0.01	500	10	0.05	15	0.2147663	0.6395	0.8132	0.7201946	0.7324938	1	0.6394984	-0.3605016	
0.8	0.01	500	10	0.1	15	0.2123824	0.6426	0.7747	0.7121212	0.7168525	1	0.6426332	-0.3573668	

Number of Trees Check

train_split	prev_filter	num.trees	max_depth	mtry_fraction	min.node.size	OOB error	CM accuracy	Sensitivity	Overall F1	AUC	OFC Train	OFC Test	Overfitting Difference	Notes
0.8	0.01	500	10	0.03	15	0.2166828	0.6426	0.8352	0.7272727	0.7144461	1	0.6426332	-0.3573668	BASELINE
0.8	0.01	1000	10	0.03	15	0.2165954	0.652	0.8407	0.7338129	0.7120799	1	0.6520376	-0.3479624	BEST IMPROVEMENT
0.8	0.01	3000	10	0.03	15	0.2165469	0.6458	0.8297	0.7277108	0.7172134	1	0.645768	-0.354232	
0.8	0.01	5000	10	0.03	15	0.216603	0.6395	0.8187	0.7215496	0.7156493	1	0.6394984	-0.3605016	
0.8	0.01	10000	10	0.03	15	0.21646	0.6458	0.8297	0.7277108	0.7140852	1	0.645768	-0.354232	
0.8	0.01	15000	10	0.03	15	0.2165773	0.6395	0.8242	0.7228916	0.7122403	1	0.6394984	-0.3605016	

Split Rule Check

train_split	prev_filter	num.trees	max_depth	mtry_fraction	min.node.size	OOB error	CM accuracy	Sensitivity	Overall F1	AUC	OFC Train	OFC Test	Overfitting Difference	Notes
0.8	0.01	3000	10	0.03	15	0.2165469	0.6458	0.8297	0.7277108	0.7172134	1	0.645768	-0.354232	BASELINE, gini
0.8	0.01	3000	10	0.03	15	0.224226	0.6207	0.8462	0.7179487	0.6836448	1	0.6206897	-0.3793103	extratrees
0.8	0.01	3000	10	0.03	15	0.2142495	0.6426	0.7967	0.7178218	0.7096735	0.9992194	0.6426332	-0.3565862	hellinger

Final Tune Check

train_split	prev_filter	num.trees	max_depth	mtry_fraction	min.node.size	OOB error	CM accuracy	Sensitivity	Overall F1	AUC	OFC Train	OFC Test	Overfitting Difference	Notes
0.8	0.01	3000	10	0.05	10	0.2139794	0.6489	0.8187	0.7268293	0.7169728	1	0.6489028	-0.3510972	Check top grid search item

69 **5 - Initial Model Results: Classifiers as HC, PA, CRC, and Other**

70

Random Forest Classifier	Included CRC Classes
HC	HC
PA	PA, PA+
CRC	CRC, CRC+
Other	Other

TABLE 8 Model classifiers for multi-class system.

71 **5-A: Summarized Results of initial model with separated CRC Progression**

72 **Classifiers**

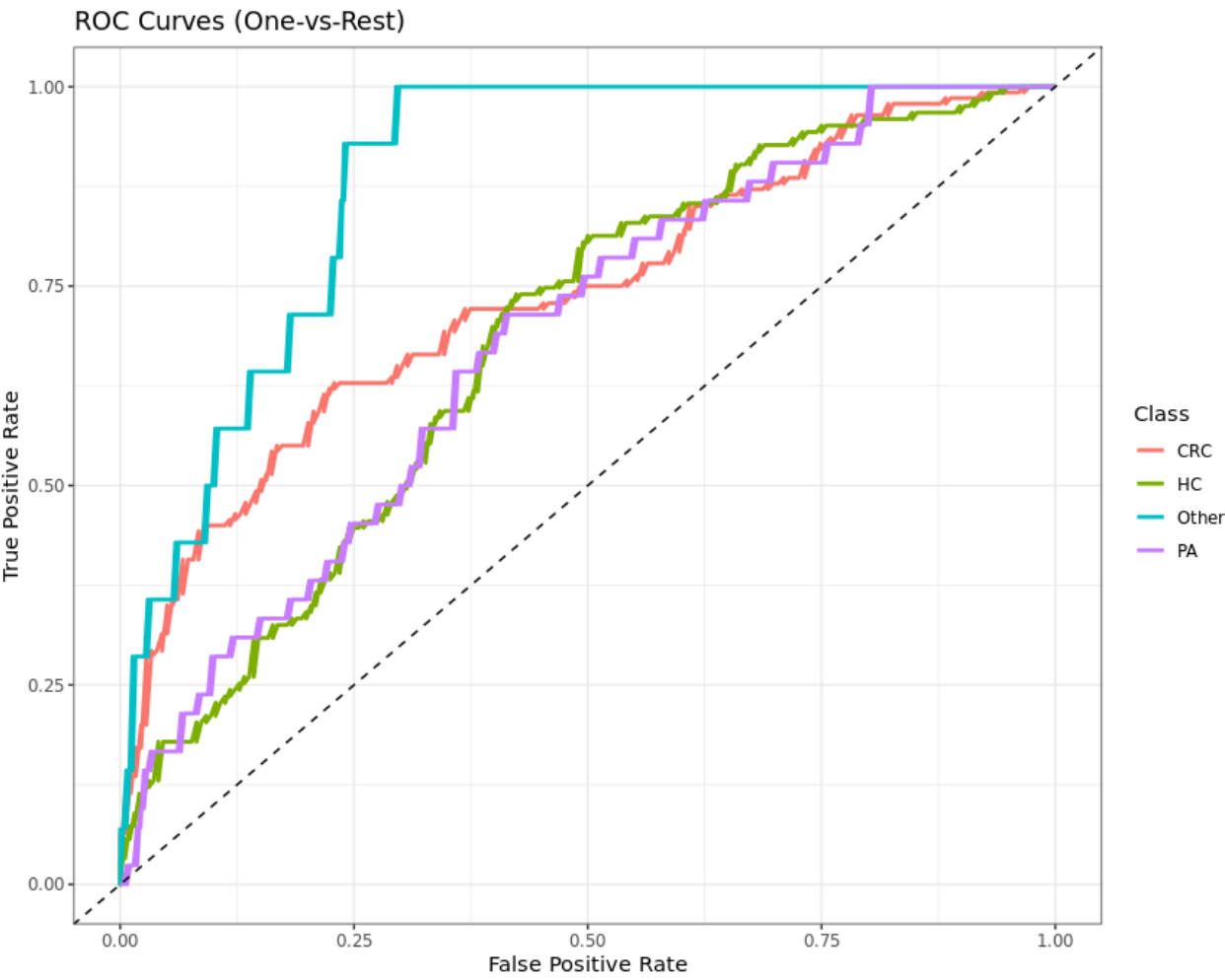
73

Metric	HC	PA	CRC	Other
n_{train}	494	170	560	57
Accuracy	0.5392	-	-	-
Balanced Accuracy	0.6029	0.4946	0.6641	0.6006
P-Value	2.014e-4	-	-	-
Kappa	0.2289	-	-	-
Sensitivity	0.5935	0.0000	0.6857	0.2143
Specificity	0.6122	0.9892	0.6425	0.9869
Positive Predictive Value	0.4899	0.0000	0.6000	0.4286
Negative Predictive Value	0.7059	0.8671	0.7233	0.9647
F1 Score	0.5368	NaN	0.6400	0.2857
Overall AUC	0.6801	0.6798	0.7362	0.8836
Overfit Check	0.2883	-	-	-

TABLE 9 Initial Model per Class Results.

$$Overfit_Check = Accuracy_{Train} - Accuracy_{Test}.$$

74 5-B: Confusion Matrix Plot



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FIG 5 Confusion Matrix for Initial Separated CRC Progression Classifiers

76 5-C: AUC-ROC Plot

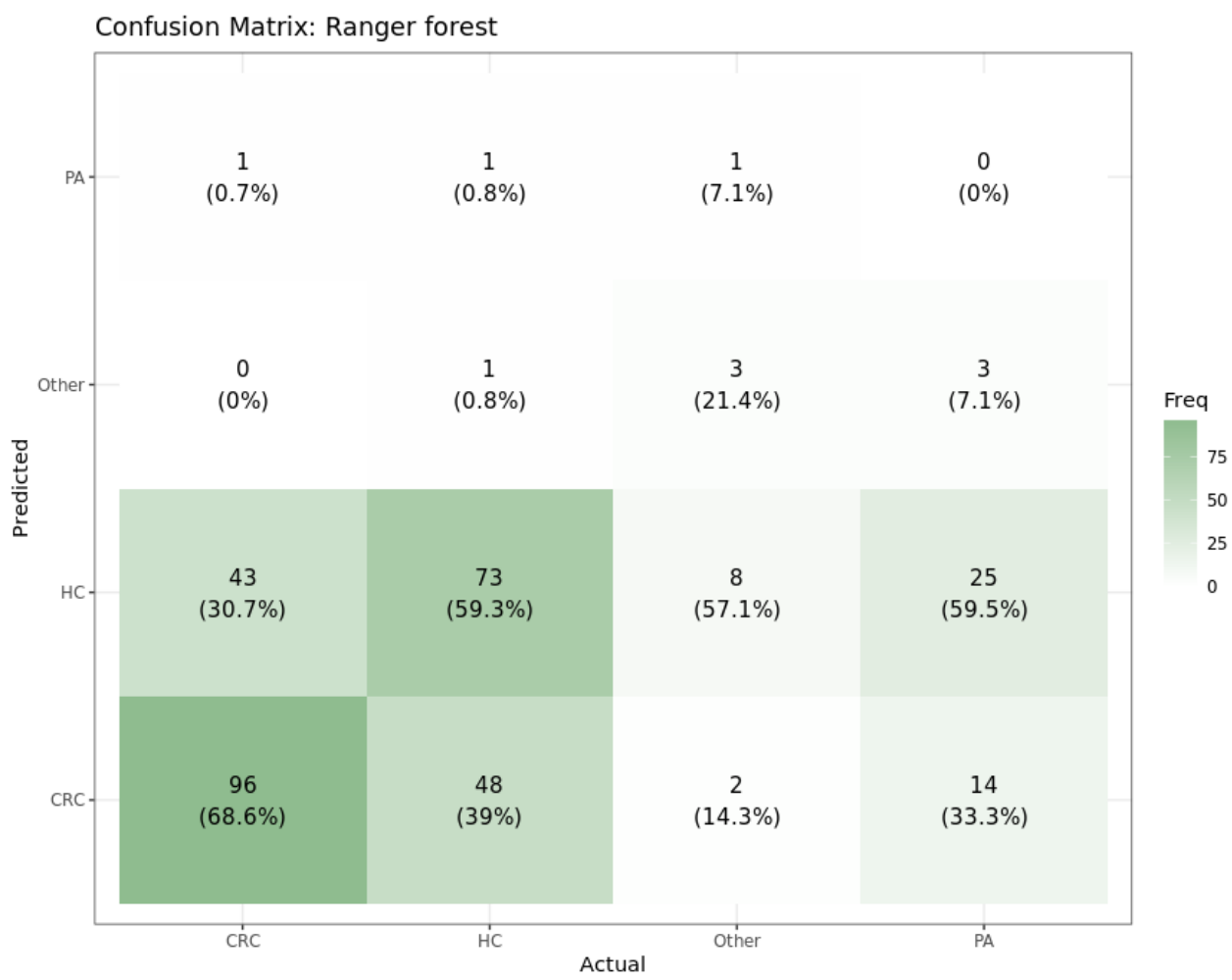


FIG 6 AUC-ROC for Initial Separated CRC Progression Classifiers

78 6 - Final Model Feature Tables

	Category	Feature	Importance
	Taxa	Gemella_morbillorum	7.3436505
	Taxa	Peptostreptococcus_stomatis	6.2277075
	Taxa	Solobacterium_moorei	4.3039837
	Pathway Abundance	PWY-6588: pyruvate fermentation to acetone	1.8075599
	Pathway Abundance	P163-PWY: L-lysine fermentation to acetate and butanoate	1.2796848
	Pathway Abundance	HISTSYN-PWY: L-histidine biosynthesis	0.8997054
	Pathway Abundance	PWY-6151: S-adenosyl-L-methionine cycle I	0.8152862
	Pathway Abundance	GLYCOGENSYNTH-PWY: glycogen biosynthesis I (from ADP-D-Glucose)	0.8015019
	Taxa	Porphyromonas_asaccharolytica	0.7978326
79	Pathway Abundance	PWY-7392: taxadiene biosynthesis (engineered)	0.6975176
	Pathway Coverage	HSERMETANA-PWY: L-methionine biosynthesis III	0.5835407
	Pathway Abundance	REDCITCYC: TCA cycle VIII (helicobacter)	0.545926
	Pathway Coverage	PWY-6936: seleno-amino acid biosynthesis	0.4954304
	Pathway Abundance	PWY4LZ-257: superpathway of fermentation (Chlamydomonas reinhardtii)	0.4836196
	Pathway Abundance	PWY-7229: superpathway of adenosine nucleotides de novo biosynthesis I	0.4652781
	Pathway Abundance	PWY-5138: unsaturated, even numbered fatty acid and beta;-oxidation	0.4094792
	Pathway Coverage	ILEUSYN-PWY: L-isoleucine biosynthesis I (from threonine)	0.4070427
	Pathway Coverage	PWY-6151: S-adenosyl-L-methionine cycle I	0.4026092
	Pathway Abundance	DTDPRHAMSYN-PWY: dTDP-L-rhamnose biosynthesis I	0.3939978
	Pathway Abundance	PWY-6892: thiazole biosynthesis I (E. coli)	0.3851618

TABLE 10 Top 20 features from 80/20 split RF model.

80 **7 - LODO AUC-ROC Curve**

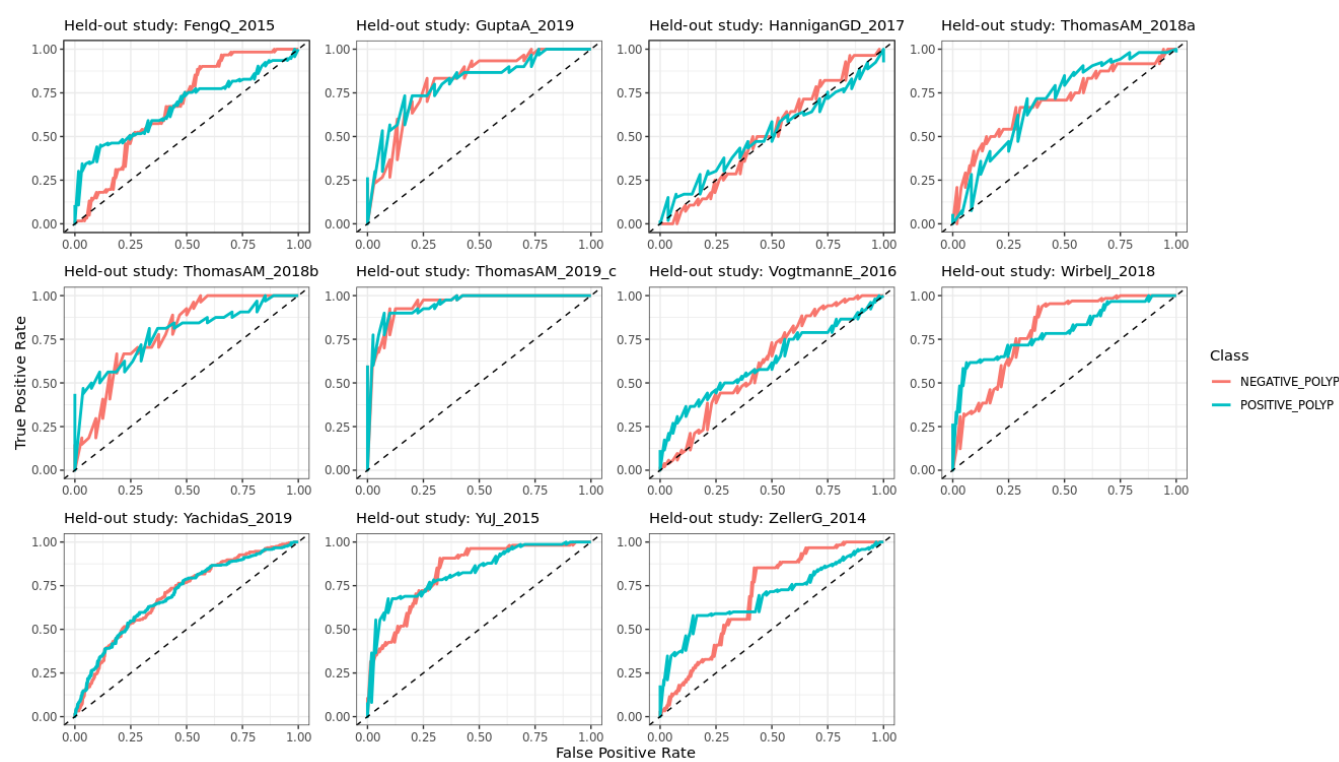


FIG 7 AUC-ROC Curve for each held-out study.

82 **8 - Relevant Links**

83 **8-A: GitHub for Script Availability**

84 Link to our project GitHub Repository: <https://github.com/chejj/Therayess>

85 **8-B: cMD3 Data Set**

86 Link to the cMD3 data set Bioconductor Page:

87 <https://bioconductor.org/packages/3.16/data/experiment/html/curatedMetagenomicData.html>